Kyubin James Kwon

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EDUCATION

University of California, Santa Barbara 2021–2027 (expected)
Ph.D. Physics (expected)
M.A. Physics 2024
Advisor: Prof. Tejaswi Venumadhav

University of California, Berkeley 2021
High Distinction, B.A. *Honors* Astrophysics; B.A. Physics
Thesis: *deepCR on ACS/WFC: Cosmic-Ray Rejection for HST ACS/WFC Photometry*
Advisor: Prof. Joshua Bloom

RESEARCH INTERESTS

Tidal interactions, neutron stars, gravitational waves, general relativity, asteroseismology, statistical methods

INVITED TALKS & SEMINARS

University of Birmingham Astrophysics Seminar, *Nonlinear Tides in Coalescing BNS* 2025
ICTS Astrophysics & Relativity Seminar, *Nonlinear Tides in Coalescing Neutron Star Binaries* 2024
Montana State University Gravity Group Seminar, *Nonlinear Tidal Effects on BNS Inspiral* 2024

CONTRIBUTED TALKS & POSTERS

APS Global Physics Summit, *Resonance Locking of g-Modes in Coalescing Neutron Star Binaries* 2026
APS Global Physics Summit, *General Relativistic Triplet Interaction of Oscillation Eigenmodes in Deformed Neutron Stars*

GWPAW25, *Resonance Locking of g-Modes in Coalescing Neutron Star Binaries* 2025
The 41st Pacific Coast Gravity Meeting, *Resonance Locking of g-Modes in Neutron Star Binaries*
APS Global Physics Summit, *Resonance Locking of g-Modes in Coalescing Neutron Star Binaries*

UCSB AstroLunch, *Resonance Locking of g-Modes in Coalescing Neutron Star Binaries* 2024
GWPAW24, *Nonlinear Resonance Locking in Coalescing Binary Neutron Stars*
SBI for Galaxy Evolution, *Normalizing Flow to Model the Kinematics of Galaxies in Halos*
The 40th Pacific Coast Gravity Meeting, *Tidal pg-instability in Coalescing Binary Neutron Stars*

UCSB AstroTea, *Neural Stellar Population Synthesis Emulator for Probabilistic VAC* 2023

UCSB Astro Lunch, *The Probabilistic Value-Added Catalog for Bright Galaxy Survey* 2022
The 54th Sierra Conference, *The Probabilistic Value-Added BGS Catalog*

AAS237, *deepCR on ACS/WFC: Cosmic-Ray Rejection Model for HST ACS/WFC Photometry* 2021

DESI Research Forum, *GQP Mock Challenge for Probabilistic Value-Added BGS Catalog* 2020

AWARDS, HONORS & GRANTS	Individualized Professional Skills Grant, UCSB	2026
	Academic Senate Doctoral Student Travel Grant, UCSB	
	DGRAV Student Travel Award, APS	
	Academic Senate Doctoral Student Travel Grant, UCSB	2025
	Graduate Student Association Travel Grant, UCSB	
	Phi Beta Kappa, UC Berkeley	2021
	Chambliss Astronomy Award Honorable Mention, AAS 237	
	Dean's List, College of Letters & Science, UC Berkeley	
RESEARCH EXPERIENCE	Academic Visitor, *University of Birmingham, United Kingdom*	2025
	Academic Visitor, *International Centre for Theoretical Sciences, India*	2023, 2024
	Graduate Student Researcher, *UC Santa Barbara, USA*	2022–
	Research Intern, *NASA Ames & KARI*[1], *USA*	Postponed
	Undergraduate Research Apprentice Program, *UC Berkeley, USA*	2019–2021
TEACHING & MENTORING	Course Reader (Statistical Mechanics), UCSB	2026
	Instructional Laboratory Group, UCSB	2025
	Lead Teaching Assistant (Physics Laboratory), UCSB	2022
	Teaching Assistant (Physics Laboratory), UCSB	2021
	Course Reader (Relativistic Astrophysics and Cosmology), UC Berkeley	2021
	Course Reader (Stellar Physics), UC Berkeley	2020
	Math Tutor, Diablo Valley College	2018–2019

TECHNICAL SKILLS

Languages: Python, C++, Shell scripting
Methods: High-Performance Computing (HPC); Parallel Computing (MPI, OpenMP); Machine Learning (TensorFlow, PyTorch)
HPC Machines: UCSB HPC Cluster; National Energy Research Scientific Computing Center–Cori (2M+ CPU hours); Exalearn (Lawrence Berkeley National Laboratory)

SERVICE & OUTREACH	Referee, ApJS	2026
	UCSB AstroTea Organizer	2023–
	Invited Writer, Astrobites	2021
	Language Mentor for War-Impacted Ukrainian Students	2023–2026
	Co-chair of Astrophysics Departmental Undergraduate Task Force, UC Berkeley	2020–2021

- Astrophysics Finals Workshop Facilitator
- Astrophysics Launch Day Orientation Facilitator
- Transfer Exploration Day Panel
- Town Hall Astro Social Panel
- Astro Summer Social Panel

PUBLICATIONS & PREPRINTS

†: first author; bold ***et al.***: name among the omitted co-authors. NASA/ADS lists 16 entries (8 refereed); 215 citations, h-index 9 (ADS, September 2026).

Neutron-Star Tides & Gravitational Waves

Kwon, K.J.†, *et al.* (in prep), *Nonlinear Perturbations of Relativistic Stars*

[1] Korea Aerospace Research Institute; the internship was postponed after the NASA Ames Center closed due to COVID-19.

Kwon, K.J.[†], *et al.* (in prep), *Adiabatic Nonlinear Oscillations of Superfluid Non-Relativistic Stars*

Yu, H., ***et al.*** 2026, arXiv e-prints, doi: 10.48550/arXiv.2607.07943, *Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions*

Hegade K.R., A., **Kwon, K.J.**, Venumadhav, T., Yu, H., & Yunes, N. 2026, arXiv e-prints, doi: 10.48550/arXiv.2605.08569, *The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response*

Hegade K.R., A., **Kwon, K.J.**, Venumadhav, T., Yu, H., & Yunes, N. 2026, PRL, 136, 071401, doi: 10.1103/1wdp-6x27, *Relativistic and Dynamical Love Numbers*

Kwon, K.J.[†], Yu, H., & Venumadhav, T. 2025, arXiv e-prints, doi: 10.48550/arXiv.2503.11837, *Resonance locking: radian-level phase shifts due to nonlinear hydrodynamics of g-modes in merging neutron star binaries*

Kwon, K.J.[†], Yu, H., & Venumadhav, T. 2024, arXiv e-prints, doi: 10.48550/arXiv.2410.03831, *Resonance Locking of Anharmonic g-Modes in Coalescing Neutron Star Binaries*

Yu, H., Weinberg, N.N., Arras, P., **Kwon, J.**, & Venumadhav, T. 2023, Monthly Notices of the Royal Astronomical Society, 519, 4325, doi: 10.1093/mnras/stac3614, *Beyond the linear tide: impact of the non-linear tidal response of neutron stars on gravitational waveforms from binary inspirals*

Statistical Methods in Astrophysics & Cosmology

Agarwal, S., ***et al.*** 2026, ApJ, 1001, 162, doi: 10.3847/1538-4357/ae4a9a, *DESI Strong Lens Foundry. III. Keck Spectroscopy for Strong Lenses Discovered Using Residual Neural Networks*

Huang, X., ***et al.*** 2026, ApJ, 998, 69, doi: 10.3847/1538-4357/ae22d2, *DESI Strong Lens Foundry. I. HST Observations and Modeling with GIGA-Lens*

Huang, X., ***et al.*** 2025, arXiv e-prints, doi: 10.48550/arXiv.2509.18089, *DESI Strong Lens Foundry II: DESI Spectroscopy for Strong Lens Candidates*

Kwon, K.J.[†], & Hahn, C. 2024, ApJ, 976, 76, doi: 10.3847/1538-4357/ad8442, *Investigating the Kinematics of Central and Satellite Galaxies Using Normalizing Flows*

Storfer, C., ***et al.*** 2024, ApJS, 274, 16, doi: 10.3847/1538-4365/ad527e, *New Strong Gravitational Lenses from the DESI Legacy Imaging Surveys Data Release 9*

Kwon, K.J.[†], Hahn, C., & Alsing, J. 2023, ApJS, 265, 23, doi: 10.3847/1538-4365/acba14, *Neural Stellar Population Synthesis Emulator for the DESI PROVABGS*

Hahn, C., **Kwon, K.J.**, *et al.* 2023, ApJ, 945, 16, doi: 10.3847/1538-4357/ac8983, *The DESI PRObabilistic Value-added Bright Galaxy Survey (PROVABGS) Mock Challenge*

NON-REFEREED PUBLICATIONS

Lovell, C.C., ***et al.*** 2023, in Machine Learning for Astrophysics, 21, doi: 10.48550/arXiv.2307.06967, *A Hierarchy of Normalizing Flows for Modelling the Galaxy-Halo Relationship* (workshop proceedings)

Kwon, K.J.[†], Zhang, K., & Bloom, J.S. 2021b, RNAAS, 5, 98, doi: 10.3847/2515-5172/abf6c8, *deepCR on ACS/WFC: Cosmic-Ray Rejection for HST ACS/WFC Photometry* (research note)

Kwon, K.J.[†], Zhang, K., & Bloom, J.S. 2021a, in AAS Meeting Abstracts, Vol. 237, AAS Meeting Abstracts, 350.02, *deepCR-ACS/WFC: Cosmic-Ray Rejection Model for HST ACS/WFC Photometry* (meeting abstract)

REFERENCES

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